

Solution Requirements

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what the Human Development Index Prediction system should do and how well it should perform.

- **Functional Requirements (FR):** Define the specific features, functions, and services the HDI Prediction System must provide.
- **Non-Functional Requirements (NFR):** Define the quality attributes such as performance, security, reliability, and usability.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	User Input	Allow users to enter HDI indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita.	High
2	Data Validation	Validate user inputs to ensure all values are complete, numeric, and within valid ranges before prediction.	High

3	Machine Learning Prediction	Load the trained Linear Regression model and generate the predicted HDI score/category based on user inputs.	High
4	Dataset Management	Store and manage the HDI dataset used for model training and prediction.	Medium
5	Result Display	Display the predicted HDI score/category clearly along with relevant insights to the user.	High
6	Visualization	Present graphical analysis of HDI indicators using charts and plots for better understanding.	Medium
7	Model Integration	Integrate the trained machine learning model with the Flask web application for real-time predictions.	High
8	<i>Other</i>	Support future model retraining and updates using newly available HDI datasets.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate HDI predictions quickly after user input submission.	Prediction generated within 2 seconds.
2	Scalability	The application should efficiently handle multiple prediction requests from different users.	Supports 100+ concurrent prediction requests.
3	Security & Data Privacy	Protect user inputs and application resources from unauthorized access.	Secure input validation and protected application environment.
4	Reliability & Availability	The system should consistently produce accurate predictions with minimal downtime.	99% application availability during normal usage.
5	Usability & Accessibility	The application should provide a simple, intuitive, and responsive interface for users.	User-friendly interface with easy navigation across devices
6	Maintainability	The system should allow easy updates to datasets, machine learning models, and application components.	Modular code structure enabling efficient maintenance and future enhancements.